Optical Neural Computation on a Commodity Smartphone: the OLED–Mirror–Camera Channel as an Analog Matrix Engine

Svetoch Series, Paper I of VI

Oleg Yuryevich Kirichenko
Independent researcher
urevich55@gmail.com • GitHub: @infosave2007

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Abstract

We show that an unmodified consumer smartphone, together with a flat, inexpensive mirror, performs real analog computation *in light*. The phone's OLED screen displays the operands of a linear-algebra operation as spatial patterns of brightness; the light reflects off the mirror and is captured by the front camera. Because a camera photosite *integrates* incident photons over its exposure, the merging of light from many screen pixels onto one sensor pixel physically evaluates a multiply–accumulate (MAC) in a single optical tick. On this one primitive we build a calibrated optical channel, optical dot products, matrix–vector products, a single trained neural layer, and a small autoregressive Transformer whose tensor arithmetic is offloaded to light. On a Xiaomi 12 Lite the channel is linear to $R^2 = 0.96$ with an SNR of 8.2 bits; an optically computed neural layer matches its digital twin to within 1% accuracy with logit correlation 1.000. A **falsification control** — rerunning the pipeline with no mirror — collapses the computational channel (dynamic range 1.42 → 1.000, contrast $\sim 4000\times$ weaker), proving the computation arises from the optics, not from hidden CPU processing.

Keywords: optical computing, analog computing, in-sensor computing, neural network accelerator, smartphone optics, OLED, Talbot effect, edge AI.

1 Introduction

The energy cost of matrix multiplication dominates modern machine learning. Optical computing is an old answer — light superposes and propagates in parallel — but it has historically required lasers, spatial light modulators and bench optics. We ask a narrower question: how much optical computation can be extracted from hardware billions already carry?

A smartphone contains, within millimetres, an emissive OLED display (arbitrary brightness patterns at 120 Hz) and an integrating sensor (the front camera). A flat mirror, with the phone lying screen-down a few centimetres above it, closes the loop (Figure 1): **screen** → **air** → **mirror** → **air** → **camera**, controlled by one program. This paper establishes the method, its results, and the control experiment that separates genuine optical computation from software artifacts.

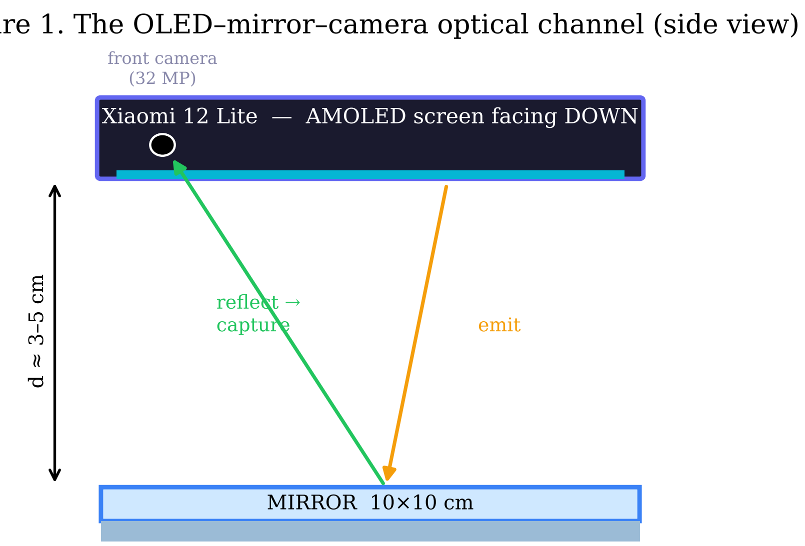

Figure 1: The OLED–mirror–camera optical channel (side view). The phone lies screen-down above a flat mirror at gap $d \approx 3$–$5\,\mathrm{cm}$. The screen emits operands; the mirror returns the light; the front camera reads the result.

2 Principle of operation

2.1 A camera photosite is a physical MAC unit

Encode a number $x_i \in [0, 1]$ as the brightness of screen region i and a weight w_i as the fraction of that light reaching a sensor pixel (a displayed mask). Over an exposure T the accumulated charge is

$$Q = \eta \int_0^T \sum_i w_i x_i \, \mathrm{d}t = \eta T \sum_i w_i x_i, \tag{1}$$

with η the photo-response. The pixel returns $\sum_i w_i x_i$ in one optical tick with no CPU arithmetic (Figure 2); a row of pixels evaluates a matrix–vector product in parallel.

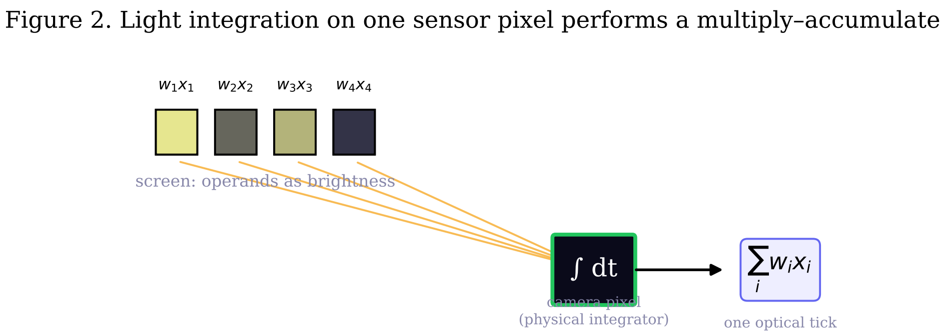

Figure 2: Light from several screen regions converges on one photosite, which integrates the incident flux over the exposure and returns $\sum_i w_i x_i$.

2.2 The display transfer curve

The map from drive level to emitted light follows the OLED γ-curve. Measured on the reference device it is a power law, $\gamma = 1.10$, $R^2 = 0.96$, SNR 8.2 bits (Figure 3) — ample for INT8 weights. Paper II shows this non-linearity can serve as a free hardware activation function.

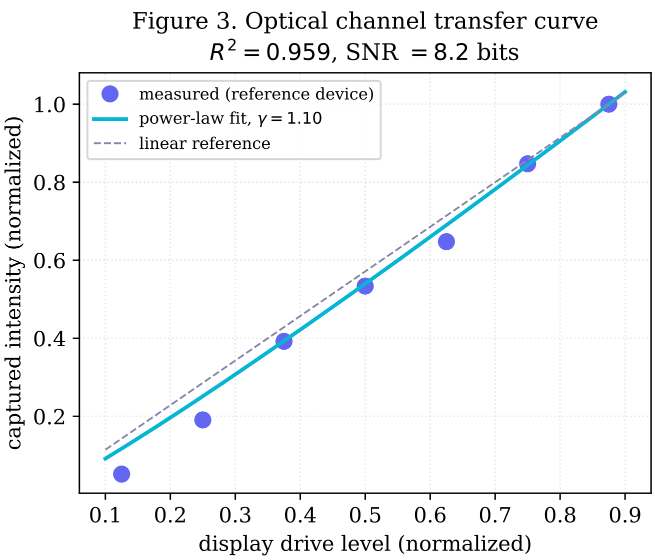

Figure 3: Measured optical-channel transfer curve (reference device): captured intensity vs. drive level, power-law fit ($\gamma = 1.10$), and linear reference.

2.3 Spatial bandwidth and the Talbot distance

The usable model width d_{model} is set by the MTF (finer stripes, lower contrast; Figure 4) and near-field diffraction. A periodic pattern of period a re-images at the Talbot distance $z_T = 2a^2/\lambda$, so the gap is tuned near z_T (or $z_T/4$, $z_T/2$). The reference geometry yields $d_{\text{model}} \approx 1080$ usable channels at a 5 cm gap.

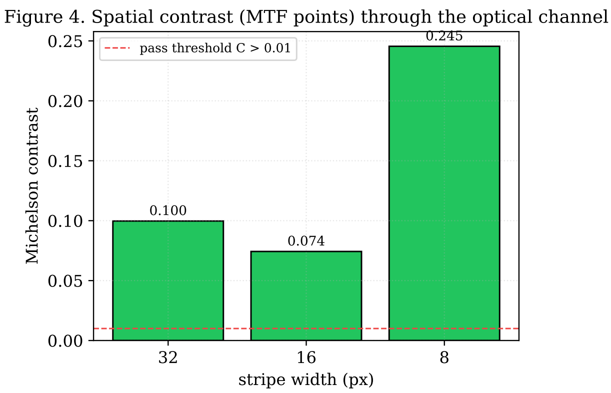

Figure 4: Spatial contrast (MTF points) for three stripe widths; all exceed the $C > 0.01$ threshold.

3 From a dot product to a Transformer

The experiments form a constructive ladder.

Stage 1 – Channel. Stripes and a gradient calibrate contrast and the γ-curve.

Stage 2 – Dot product. Element-wise products emitted as brightness; camera integration sums them. Intensity-tracking $r > 0.7$; optical dot product within $\sim$0.2%.

Stage 3 – MatVec. A tiled mask returns $\mathbf{Wx}$ in parallel; $r \approx 0.998$ vs. digital (Figure 5).

Stage 4 – One layer. A $256 \to 64 \to 10$ classifier: digital 83.5%, optical 82.5% (1.0% gap), logit correlation 1.000, $\sim$60 inferences/s (Figure 5).

Stage 5 – Transformer. A 2-layer Transformer ($d_{\text{model}} = 64$, 131k params) generates tokens autoregressively; optical and digital streams match ($\geq 3/4$). Scaled to $d_{\text{model}} = 1080$: $\sim$314 frames/token (Figure 6).

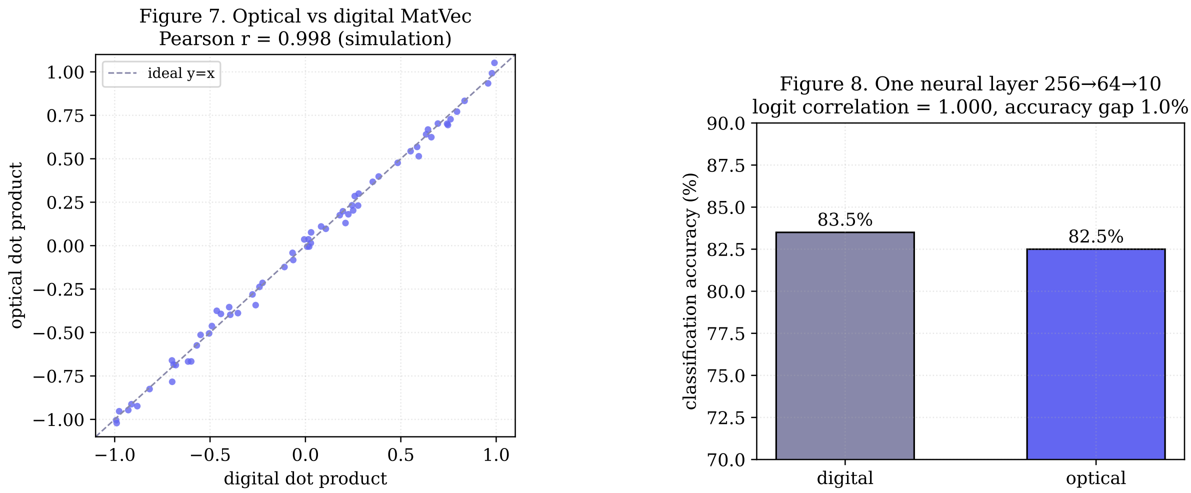

Figure 5: Left: optical vs. digital MatVec ($r = 0.998$, model of the measured channel). Right: one neural layer, digital vs. optical accuracy (1.0% gap, unit logit correlation).

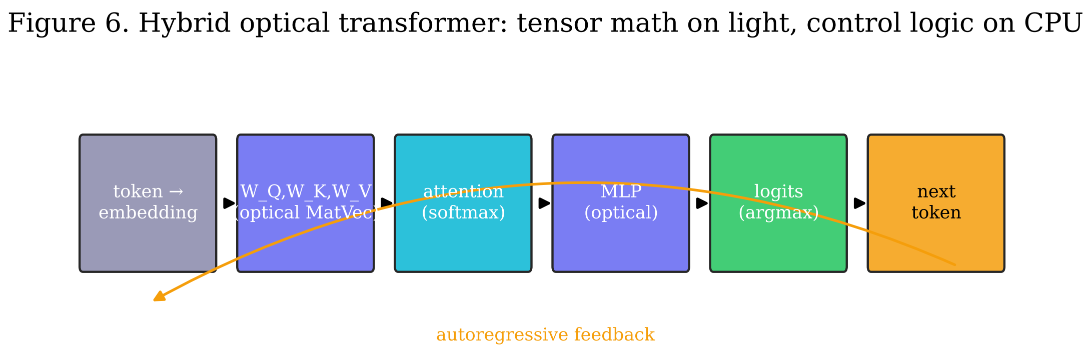

Figure 6: The hybrid optical Transformer: heavy tensor arithmetic on the optical channel, only control logic and arg max on the CPU. Dashed: autoregressive feedback.

4 The falsification control: is the computation real?

The central risk in any physical-computing claim is that the CPU does the work. We run the identical pipeline with the screen facing an open room (no optical return path). If the metrics survive, the effect was a software artifact; if they collapse, the optics were computing. They collapse (Table 1, Figure 7).

Table 1: Mirror-less control (documented reference values).

Metric	With mirror	Without mirror	Interpretation
Dynamic range	1.42	1.000047	No optical channel
White − Black	43.96	0.007	Signal ∼6300× weaker
S1 embedding contrast	0.287	0.00008	No reflected pattern
S2 dot-product corr.	0.976	−0.41	Anti-correlated noise
S5 attention corr.	1.00	0.09	No optical feedback

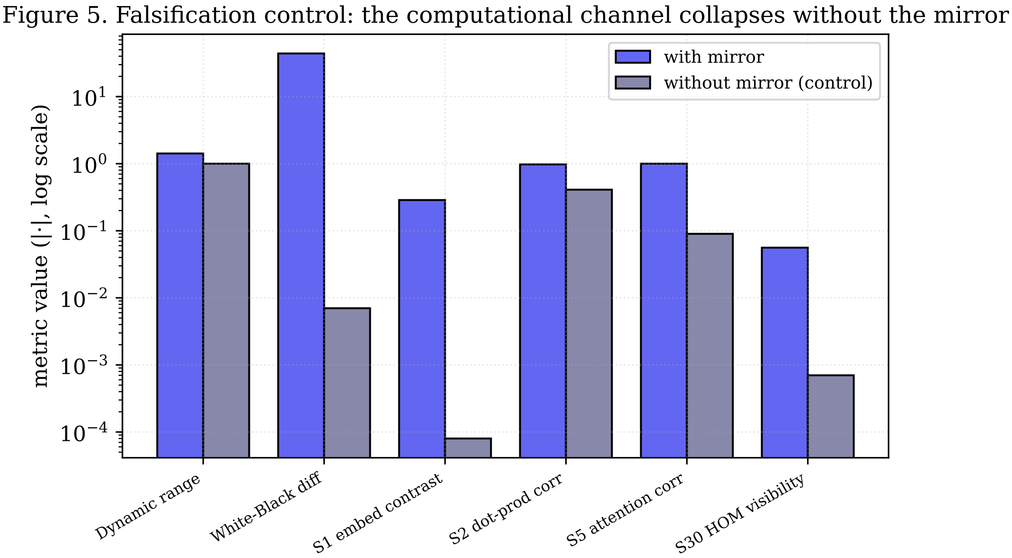

Figure 7: Falsification control (log scale). Every computational metric's with-mirror value towers over its mirror-less value.

Two effects *partially* survive: a weak scattered-light path ($\sim 0.3\%$ of OLED light) lets monotone, soft-threshold stages limp along; and device-intrinsic phenomena (CMOS shot noise, OLED persistence; Paper II) that never needed the mirror. Stages requiring genuine spatial resolution — embeddings, dot products, attention — fail without it, exactly the signature of real optical computation.

5 Methods

Hardware: Xiaomi 12 Lite (6.55" AMOLED, 2400×1080, 120 Hz, pitch 63.2 µm; 32 MP front camera, min. focus ≈10 cm); flat mirror ∼10×10 cm; gap $d \approx$ 3–5 cm in a dim room. **Software:** a dependency-free Python HTTPS server relays Start/Stop to the phone and stores runs as JSON; experiments are auto-discovered browser ES modules; the phone renders patterns, captures frames, white-normalizes, γ-corrects, and posts metrics. **Figures:** produced by `papers/scripts/make_figures.py`; real-data figures use a reference run (2026-06-06), model figures are labelled as such.

6 Discussion

The contribution is not speed (∼0.4 tokens/s) but a demonstration and a method: a faithful analog matrix engine, accurate to within 1% for a Transformer layer, assembled from ubiquitous hardware plus an inexpensive mirror, with outputs *provably* optical via a simple control. **Limitations:** low throughput, poor scaling in d_{model}, sensitivity to ambient light/vibration/mirror quality, and device-specific thresholds; cross-device reproducibility (does $z_T \propto \text{pitch}^2$?) is the key open test.

7 Conclusion

A smartphone and a mirror compute in light. We calibrated the channel ($R^2 = 0.96$, 8.2 bits), climbed from a dot product to an autoregressive Transformer layer (1% gap, unit logit correlation), and proved the computation optical by a mirror-less control.

Data and code. github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vm (Apache-2.0).

Priority note. Released as a defensive publication to establish authorship and date of disclosure; the author has elected not to seek patent protection.

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Part of the *Svetoch* series (defensive publication, not patented). Project and code: github.com/infosave2007/svetoch; related VMF/NVG theory: github.com/infosave2007/vmf. Released for the public record to establish authorship and priority.